

LITERATURE SURVEY REPORT

**VL-JEPA vs. CLIP for CCTV Event Retrieval:
Efficiency, Generalization, and Faithful Report Generation**

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Capstone Project Literature Survey

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Abstract

This report presents an expanded, structured literature survey supporting the capstone project “VL-JEPA vs. CLIP for CCTV Event Retrieval: Efficiency, Generalization, and Faithful Report Generation.” The project investigates whether pretrained VL-JEPA (Joint Embedding Predictive Architecture for Vision-Language) representations can match or exceed CLIP-based representations for semantic event retrieval in CCTV surveillance footage, how well each representation generalizes across cameras and viewpoints under limited computational resources, and whether grounding LLM-generated incident reports in retrieved visual evidence reduces hallucinated content. This revision incorporates four additional peer-reviewed sources published in 2025–2026 that directly evaluate CLIP-family and LLM-based pipelines on CCTV-style and safety-critical reporting tasks, strengthening the empirical grounding of the project’s problem statement. In total, this survey reviews thirty-three literature sources spanning eight thematic areas: (A) VL-JEPA foundations, (B) JEPA and self-supervised video representation learning, (C) CLIP and its retrieval- and anomaly-oriented derivatives, (D) CCTV and surveillance-specific vision-language systems, (E) video retrieval and retrieval-augmented generation (RAG), (F) detection, tracking and general video representation backbones, (G) LLM hallucination and faithfulness evaluation, and (H) cross-camera and cross-domain generalization. Each section synthesizes the reviewed works and identifies the specific gap that the proposed framework addresses.

Contents

Abstract	1
1 Introduction	3
2 VL-JEPA Papers (Category A)	3
3 JEPA / Video Representation Papers (Category B)	4
4 CLIP Papers (Category C)	4
5 CCTV / Surveillance Papers (Category D)	5
6 Video Retrieval & RAG Papers (Category E)	7
7 Detection, Tracking & Video Representation Papers (Category F)	7
8 LLM Hallucination & Faithfulness Papers (Category G)	8
9 Cross-Camera Generalization Papers (Category H)	9
10 Cross-Cutting Discussion and Research Gap	10
11 Consolidated Reference List	10

1 Introduction

The proliferation of CCTV infrastructure in industrial, public, and residential settings has produced volumes of video data that are impractical to monitor or search manually. Vision-language models such as CLIP have made it possible to query video archives using natural language rather than manually engineered event categories, but CLIP was designed primarily for static image-text alignment and its behaviour on video, across changing camera viewpoints, and under constrained compute budgets is not fully characterised for the surveillance domain. VL-JEPA, a more recent joint-embedding predictive architecture, offers an alternative representation strategy that avoids autoregressive generation and has reported competitive results on general video-text benchmarks, but its suitability specifically for CCTV-style event retrieval remains under-explored.

A second, largely independent problem arises once relevant footage has been retrieved: converting that evidence into a natural-language incident report using an LLM introduces a risk of hallucinated or unsupported claims, which is safety-critical in a surveillance context. Recent 2025–2026 literature — reviewed for the first time in this revision — shows that this risk is not hypothetical: comparable vision-language-plus-LLM surveillance pipelines (e.g., warehouse and narrator systems) and safety-critical summarization studies in adjacent domains (e.g., maritime accident reporting) both confirm that natural-language reporting layered on top of visual or textual evidence introduces measurable, systematically characterisable hallucination risk. This survey reviews the literature underpinning both halves of the problem — representation learning for retrieval, and faithfulness of generated reports — organised into the eight categories used for source collection, with newly incorporated sources integrated directly into the relevant categories (C, D, and G) rather than treated as a separate addendum.

2 VL-JEPA Papers (Category A)

VL-JEPA is the representation family at the centre of the project’s main hypothesis (H1, H2). These two sources define the architecture and its officially reported evaluation, and form the baseline against which the project’s own CCTV-domain experiments are compared.

S.No	Paper Title	Year	Application	Link
1	VL-JEPA: Joint Embedding Predictive Architecture for Vision-Language	2025	Vision-language retrieval, video understanding, VQA, semantic retrieval	https://arxiv.org/abs/2512.10942
2	VL-JEPA (ICLR / OpenReview Version)	2025	Official conference version	https://openreview.net/forum?id=tjimrqc2BUB

Synthesis. Together these establish VL-JEPA as a joint-embedding predictive approach to vision-language representation that, unlike CLIP-style contrastive image-text encoders, learns by predicting representations rather than raw pixels or tokens. The project treats the general-domain results reported here as the starting hypothesis to be re-tested under CCTV-specific conditions: lower resolution, static viewpoints, and resource-constrained inference.

3 JEPA / Video Representation Papers (Category B)

This category covers the broader JEPA family and self-supervised video encoders that VL-JEPA builds on conceptually, providing background for objective 1 (framework design) and objective 4 (computational analysis).

S.No	Paper Title	Year	Application	Link
3	V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning	2025	Video understanding, robotics, action prediction	https://arxiv.org/abs/2506.09985
4	V-JEPA 2.1	2026	Dense video representation learning	https://arxiv.org/abs/2603.14482

***Synthesis.** V-JEPA and its successors demonstrate that predictive, non-generative self-supervised objectives can yield transferable video representations useful for downstream understanding and planning tasks. These works motivate treating VL-JEPA as a video-native alternative to image-pretrained CLIP encoders, and support the project’s framing that predictive embeddings may generalize differently than contrastive ones when camera viewpoint or domain shifts, which is directly relevant to hypothesis H2.*

4 CLIP Papers (Category C)

CLIP and its retrieval- and anomaly-oriented derivatives form the comparison baseline for objectives 1–3, since the project explicitly measures VL-JEPA against CLIP-family representations on retrieval accuracy, generalization, and efficiency. This revision adds a sixth source that applies a temporally-extended CLIP directly to CCTV anomaly classification, strengthening the empirical bridge between general CLIP research and the project’s own surveillance setting.

S.No	Paper Title	Year	Application	Link
5	Learning Transferable Visual Models From Natural Language Supervision (CLIP)	2021	Semantic retrieval, zero-shot learning	https://arxiv.org/abs/2103.00020
6	VideoCLIP: Contrastive Pre-training for Zero-shot Video-Text Understanding	2021	Video retrieval, video captioning	https://arxiv.org/abs/2109.14084
7	VadCLIP: Adapting CLIP for Video Anomaly Detection	2023	CCTV anomaly detection	https://arxiv.org/abs/2307.13425
8	CLIP-ReID	2022	Cross-camera person retrieval	https://arxiv.org/abs/2211.13977
9	RegionCLIP	2022	Open-vocabulary object detection	https://arxiv.org/abs/2112.09106

S.No	Paper Title	Year	Application	Link
10	Using Vision-Language Models to Detect Anomalies in CCTV Surveillance Videos	2026	Temporal CLIP + text-fusion for multi-class CCTV anomaly detection, with Sora synthetic-video augmentation	https://ieeexplore.ieee.org/document/11449611

Synthesis. The original CLIP paper established large-scale contrastive image-text pretraining as a route to zero-shot semantic retrieval, and the following sources show the surveillance-relevant lineage that follows from it: VideoCLIP extends the contrastive objective to video-text pairs; VadCLIP and RegionCLIP adapt CLIP features to anomaly detection and region-level open-vocabulary detection respectively; CLIP-ReID applies CLIP-derived embeddings to cross-camera person retrieval; and Choi et al.’s 2026 ICCE paper (newly added) goes furthest toward the project’s own setting by fusing a CLIP ViT-B/16 backbone with a lightweight temporal transformer and class-specific text embeddings, then fine-tuning the combination directly on a seven-class CCTV anomaly task (Invasion, Fall down, Fight, Gathering, Crowd, Flood, plus Normal). Reported gains over an image-level CLIP baseline (macro-F1 rising from 0.41 to 0.65; accuracy from 0.47 to 0.71) demonstrate two things directly relevant to this project: (i) plain CLIP embeddings, used image-wise without temporal modelling, perform poorly on CCTV-style multi-class events, which motivates evaluating whether VL-JEPA’s native video-predictive representation closes this gap without requiring an added temporal transformer; and (ii) rare-event classes (fight, fall) remain the hardest even after enhancement, reinforcing the project’s motivation for evaluating retrieval-style (rather than closed-set classification) approaches on CCTV footage. This body of work confirms CLIP’s flexibility across surveillance-adjacent tasks while also indicating that most gains come from task-specific temporal and semantic adaptation rather than the base encoder alone — a gap the project addresses by testing pretrained (not fine-tuned) CLIP and VL-JEPA under matched, resource-constrained conditions.

5 CCTV / Surveillance Papers (Category D)

These sources anchor the project within current CCTV-specific vision-language systems, informing the problem statement and the report-generation component (objective 5). Three sources are newly added in this revision, each demonstrating a working VLM/LLM surveillance pipeline against which the project’s own CCTV-retrieval-plus-reporting framework can be positioned.

S.No	Paper Title	Year	Application	Link
11	AssistPDA: Online Video Surveillance Assistant for Video Anomaly Prediction, Detection and Analysis	2025	AI surveillance assistant	https://arxiv.org/abs/2503.21904
12	Cerberus: Cascaded Vision-Language Models for Real-Time Video Anomaly Detection	2025	Real-time anomaly detection	https://arxiv.org/abs/2510.16290

S.No	Paper Title	Year	Application	Link
13	VERA: Verbalized Learning for Explainable Video Anomaly Detection	2025	Explainable surveillance	https://openaccess.thecvf.com/
14	The Evolution of Video Anomaly Detection: From DNN to MLLM	2025	Survey of video anomaly detection	https://arxiv.org/abs/2507.21649
15	SAFER-VLM: A Vision-Language and LLM Based Framework for Intelligent Warehouse Surveillance	2025	YOLOX + ByteTrack detection/tracking, VLM per-worker activity logging, LLM zone-wise summarization and rule-based alerting	https://ieeexplore.ieee.org/document/11396276
16	Smart Surveillance Narrator: An Intelligent Video Analytics System for Automated Time-Stamped Narration and Keyword-Based Incident Retrieval in CCTV Surveillance	2026	YOLOv8 + TCN activity recognition, template-based NLG, time-stamped keyword-searchable narration database	https://ieeexplore.ieee.org/document/11588360

Synthesis. The surveyed systems trace a shift in CCTV analytics from single-purpose deep networks toward vision-language and multimodal-LLM-based assistants (AssistPDA, Cerberus) that can explain their decisions in natural language (VERA), a trajectory summarised by the DNN-to-MLLM survey. The two newly added systems make this trajectory concrete rather than aspirational. SAFER-VLM chains a detector (YOLOX) and tracker (ByteTrack) with a VLM (VideoChat-Flash) that produces per-worker natural-language activity logs, which an LLM (Mistral-7B or Llama-3) then consolidates into zone-wise summaries and rule-based alerts (92% alert precision for restricted-zone and missing-safety-gear violations); notably, its 84% VLM activity-description accuracy and reliance on background-substitution preprocessing to disambiguate overlapping workers both illustrate a practical failure mode — VLM outputs degrade under crowding and occlusion — that a retrieval-first architecture (as opposed to always-on captioning) could partially sidestep by only invoking generation on retrieved, query-relevant clips. Smart Surveillance Narrator, in turn, demonstrates an alternative, non-LLM design point: a YOLOv8 + temporal-convolutional-network pipeline paired with template-guided (not generative) narration, achieving 91.4% event detection accuracy, 94.2% timestamp accuracy, and 93.7% keyword-search F1 with a 78% reduction in manual investigation time. Its ablation study — showing that removing NLP-enhanced indexing drops search F1 from 93.7% to 87.4%, and that removing activity recognition entirely drops it to 74.1% — is directly informative for the project’s own retrieval evaluation design, since it isolates how much of end-to-end CCTV retrieval quality is attributable to the underlying visual/activity representation versus the surrounding indexing and search machinery. Both systems, however, rely on template-based or short-context LLM summarization rather than retrieval-grounded generation, and neither reports a quantitative faithfulness or hallucination evaluation of its generated text against source footage — the specific gap that motivates this project’s H3 and its report-generation component (objective 5). Taken together with AssistPDA, Cerberus, VERA, and the DNN-to-MLLM survey, these sources confirm that natural-language grounded surveillance analysis is an active and rapidly maturing research direction, but also confirm that no reviewed system directly benchmarks a joint-embedding predictive representation like VL-JEPA against CLIP under identical CCTV retrieval conditions, nor pairs that comparison with a rigorous, source-grounded faithfulness evaluation of the resulting reports — the specific gap this project targets.

6 Video Retrieval & RAG Papers (Category E)

These works inform the retrieval-augmented report generation pipeline used to satisfy objective 5 (faithfulness of generated incident reports).

S.No	Paper Title	Year	Application	Link
17	VideoRAG: Retrieval-Augmented Generation over Video Corpus	2025	Video retrieval + report generation	https://arxiv.org/abs/2501.05874
18	VideoRAG for Extremely Long Context Videos	2025	Long CCTV video search	https://arxiv.org/abs/2502.01549
19	MissionGNN	2025	Knowledge graph from events	https://openaccess.thecvf.com/
20	GenDFIR	2024	Timeline generation using RAG	https://arxiv.org/abs/2409.02572

Synthesis. VideoRAG and its long-context variant show that retrieval-augmented generation can ground LLM outputs in retrieved video segments rather than free generation, directly relevant to reducing hallucination in CCTV incident reports. MissionGNN’s knowledge-graph construction from events and GenDFIR’s RAG-based timeline generation further indicate that structuring retrieved evidence (as events, graphs, or timelines) before text generation is a common strategy for improving faithfulness — a design principle the project adopts in evaluating report faithfulness against retrieved visual evidence (hypothesis H3), and one that finds independent empirical support in the newly reviewed maritime hallucination study (Category G), which found that full-document RAG grounding materially improved fact-grounding scores over partial-segment retrieval across all tested LLMs.

7 Detection, Tracking & Video Representation Papers (Category F)

These establish the supporting computer-vision building blocks (detection, tracking, action and pose recognition) referenced under Subject Knowledge and Technical Requirements, and now also underpin two of the newly reviewed CCTV systems in Category D.

S.No	Paper Title	Year	Application	Link
21	VideoMAE	2022	Self-supervised video learning	https://arxiv.org/abs/2203.12602
22	SlowFast Networks for Video Recognition	2019	Action recognition	https://arxiv.org/abs/1812.03982
23	ByteTrack	2021	Multi-object tracking	https://arxiv.org/abs/2110.06864

S.No	Paper Title	Year	Application	Link
24	DeepSORT	2017	Object tracking	https://arxiv.org/abs/1703.07402
25	ST-GCN	2018	Human pose & fall detection	https://arxiv.org/abs/1801.07455

Synthesis. VideoMAE and SlowFast represent masked-autoencoding and two-stream convolutional approaches to general video representation learning, respectively, and serve as older, non-vision-language baselines for comparison. ByteTrack and DeepSORT are established multi-object tracking algorithms that would be required if the framework is extended from clip-level retrieval to persistent object-level surveillance analytics; notably, ByteTrack is precisely the tracker used in SAFER-VLM (Category D) to preserve worker identity across frames prior to VLM-based activity logging, confirming that this detection-tracking layer is not merely a textbook baseline but an active component of state-of-the-art CCTV pipelines. ST-GCN’s skeleton-based graph convolutional approach remains relevant to pose- and fall-detection use cases, echoing the “fall down” anomaly class evaluated in the newly reviewed CLIP-anomaly paper (Category C). These are included as supporting infrastructure rather than as direct competitors to VL-JEPA or CLIP in the retrieval comparison.

8 LLM Hallucination & Faithfulness Papers (Category G)

These papers provide the evaluation methodology for hypothesis H3 — measuring whether grounding reduces hallucination in generated incident reports. This revision adds a substantial, directly relevant empirical study of hallucination in LLM-generated summaries of safety-critical technical reports, which the project can adapt almost directly for CCTV incident-report evaluation.

S.No	Paper Title	Year	Application	Link
26	SelfCheckGPT	2023	Hallucination detection	https://arxiv.org/abs/2303.08896
27	Hallucination in Large Language Models: A Survey	2023	LLM hallucination survey	https://arxiv.org/abs/2311.05232
28	Faithfulness in Natural Language Generation	2022	Report faithfulness evaluation	https://arxiv.org/abs/2207.07408
29	To Err Are LLMs: Profiling Hallucinations in Generated Summaries of Maritime Accident Reports	2026	Fine-grained hallucination taxonomy, MARHAS dataset, automated factuality/grounding benchmark, and human-vs-LLM Pareto analysis for safety-critical report summarization	https://ieeexplore.ieee.org/document/11515092

Synthesis. SelfCheckGPT proposes a sampling-based, reference-free method for detecting hallucinated content in LLM outputs, while the accompanying survey categorises hallucination types and mitigation strategies across the broader LLM literature. The faithfulness paper addresses the related but distinct problem of evaluating whether generated text is supported by its source content in natural language generation tasks. Yuan et al.’s 2026 maritime study (newly added) operationalises these ideas in a domain that is structurally close to CCTV incident reporting: both settings require converting a factual, chronologically ordered source record (a full accident report; retrieved surveillance evidence) into a concise natural-language summary without introducing unsupported dates, entities, numbers, or causal claims. Three of its findings transfer directly to this project’s design of hypothesis H3 experiments. First, its five-category intrinsic-hallucination taxonomy — unit-conversion errors, date/time errors, number/named-entity errors, misgeneration (fabricated or contradicted content), and under/over-generation (omitted or invented detail) — is domain-agnostic and can be adapted essentially unchanged to annotate CCTV incident reports for the presence of fabricated timestamps, misidentified entities, or invented event sequences. Second, its finding that standard automatic quality metrics (BLEU, METEOR, ROUGE, BERTScore) show no significant correlation with hallucination frequency (Pearson tests, $p > 0.05$ in nearly all model/prompt-mode combinations) is a direct warning against relying on such metrics alone to validate the project’s own report-generation component, and argues for adopting a grounding-score-style automated benchmark (modelled on the paper’s FACTS-inspired grounding score, $\text{Score} = N_{\text{supported}} / (N_{\text{supported}} + N_{\text{unsupported}})$) alongside human annotation. Third, the paper’s finding that full-document context consistently improves fact-grounding scores over fragmented, partial-segment context (e.g., ChatGPT’s grounding score rising from 0.51 to 0.73) — and that RAG-based generation was not uniformly superior to few-shot prompting once intrinsic model capability was accounted for — provides an important, more nuanced counterpoint to the RAG-oriented papers in Category E, and should inform how aggressively the project chunks retrieved CCTV evidence before passing it to a report-generation LLM. Together, these four sources give the project a methodological basis for scoring CCTV incident reports for unsupported claims relative to retrieved visual evidence, rather than relying on ad hoc manual review or automatic text-similarity metrics alone.

9 Cross-Camera Generalization Papers (Category H)

These works directly support objective 3 and hypothesis H2, which concern generalization of retrieval performance across cameras and viewpoints.

S.No	Paper Title	Year	Application	Link
30	CLIP-ReID	2022	Cross-camera retrieval	https://arxiv.org/abs/2211.13977
31	TransReID	2021	Transformer-based person re-identification	https://arxiv.org/abs/2102.04378
32	Deep Learning for Person Re-identification: A Survey	2020	Cross-camera matching survey	https://arxiv.org/abs/2001.04193

Synthesis. *CLIP-ReID and TransReID represent two approaches — vision-language adaptation and pure transformer architectures, respectively — to the cross-camera person re-identification problem, where the same subject must be matched across non-overlapping camera views. The accompanying survey situates both within the broader re-identification literature. While this project’s retrieval task (semantic event retrieval, not identity matching) differs from re-identification, the cross-camera generalization challenge is structurally similar, and these papers inform the evaluation protocol used to test VL-JEPA and CLIP across different cameras or video sources.*

10 Cross-Cutting Discussion and Research Gap

Reviewing the eight categories together, now inclusive of the four newly added 2025–2026 sources, sharpens rather than changes the picture established in the original survey. VL-JEPA (Category A, B) and CLIP (Category C) have each been evaluated extensively on general-purpose video-text benchmarks and, in CLIP’s case, adapted successfully to several surveillance-adjacent tasks such as anomaly detection (VadCLIP, and now the temporally-enhanced CLIP of Choi et al.) and cross-camera retrieval (CLIP-ReID). Modern CCTV-specific systems (Category D) are increasingly built on vision-language and multimodal-LLM foundations, and the newly reviewed SAFER-VLM and Smart Surveillance Narrator systems confirm that end-to-end detection-to-narration pipelines are already operational at conference-paper maturity, achieving strong quantitative results (e.g., 90%+ event/activity accuracy, 90%+ alert precision) on their respective CCTV and warehouse datasets.

However, three gaps persist even after this expanded review, and each maps onto one of the project’s hypotheses. First, no reviewed source — including the newly added CCTV-adjacent systems — directly benchmarks pretrained VL-JEPA against pretrained CLIP specifically for CCTV-style event retrieval under resource-constrained inference (H1, H2): the newly reviewed CLIP-anomaly paper and SAFER-VLM both use CLIP or CLIP-derived VLMs as one fixed architectural choice rather than as one arm of a controlled comparison against a joint-embedding predictive alternative. Second, none of the reviewed CCTV systems pairs its retrieval or activity-recognition component with a rigorous, source-grounded faithfulness evaluation of its generated reports; SAFER-VLM and Smart Surveillance Narrator both generate natural-language output (LLM summaries and templated narrations respectively) without measuring hallucination rate against the underlying video evidence, leaving H3 empirically untested in the CCTV domain specifically. Third, and newly clarified by the maritime hallucination study, standard automatic text-quality metrics are demonstrably unreliable proxies for factual faithfulness in exactly this class of report-generation task, which means the project cannot substitute BLEU/ROUGE-style scoring for a dedicated, annotation-based or grounding-score-based faithfulness evaluation when validating H3.

This is the gap the capstone project is designed to fill: a lightweight, reproducible framework that (i) compares pretrained VL-JEPA and CLIP representations on CCTV retrieval metrics, (ii) tests generalization across cameras/viewpoints, (iii) profiles computational cost under constrained resources, and (iv) evaluates the faithfulness of LLM-generated incident reports grounded in the retrieved evidence — using an evaluation methodology directly informed by the fine-grained, domain-adapted hallucination taxonomy and grounding-score approach demonstrated in the newly reviewed maritime accident-report study, applied for the first time to CCTV-style incident reporting.

11 Consolidated Reference List

A. VL-JEPA Papers

- [1] VL-JEPA: Joint Embedding Predictive Architecture for Vision-Language (2025). <https://arxiv.org/abs/2512.10942>
- [2] VL-JEPA (ICLR / OpenReview Version) (2025). <https://openreview.net/forum?id=tjimrqc2BUB>

B. JEPA / Video Representation Papers

- [3] V-JEPA 2: Self-Supervised Video Models Enable Understanding, Prediction and Planning (2025). <https://arxiv.org/abs/2506.09985>
- [4] V-JEPA 2.1 (2026). <https://arxiv.org/abs/2603.14482>

C. CLIP Papers

- [5] Learning Transferable Visual Models From Natural Language Supervision (CLIP) (2021). <https://arxiv.org/abs/2103.00020>
- [6] VideoCLIP: Contrastive Pre-training for Zero-shot Video-Text Understanding (2021). <https://arxiv.org/abs/2109.14084>
- [7] VadCLIP: Adapting CLIP for Video Anomaly Detection (2023). <https://arxiv.org/abs/2307.13425>
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D. CCTV / Surveillance Papers

- [11] AssistPDA: Online Video Surveillance Assistant for Video Anomaly Prediction, Detection and Analysis (2025). <https://arxiv.org/abs/2503.21904>
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- [16] Lavanya S, Prabhavathi M, Nafisa F, and Suchitra S, “Smart Surveillance Narrator: An Intelligent Video Analytics System for Automated Time-Stamped Narration and Keyword-Based Incident Retrieval in CCTV Surveillance,” in *Proc. 2026 Int. Conf. System, Computation, Automation and Networking (ICSCAN)*, 2026, doi: 10.1109/ICSCAN66520.2026.11588360.

E. Video Retrieval & RAG Papers

- [17] VideoRAG: Retrieval-Augmented Generation over Video Corpus (2025). <https://arxiv.org/abs/2501.05874>
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- [20] GenDFIR (2024). <https://arxiv.org/abs/2409.02572>

F. Detection, Tracking & Video Representation Papers

- [21] VideoMAE (2022). <https://arxiv.org/abs/2203.12602>
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- [23] ByteTrack (2021). <https://arxiv.org/abs/2110.06864>
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- [25] ST-GCN (2018). <https://arxiv.org/abs/1801.07455>

G. LLM Hallucination & Faithfulness Papers

- [26] SelfCheckGPT (2023). <https://arxiv.org/abs/2303.08896>
- [27] Hallucination in Large Language Models: A Survey (2023). <https://arxiv.org/abs/2311.05232>
- [28] Faithfulness in Natural Language Generation (2022). <https://arxiv.org/abs/2207.07408>
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H. Cross-Camera Generalization Papers

- [30] CLIP-ReID (2022). <https://arxiv.org/abs/2211.13977>
- [31] TransReID (2021). <https://arxiv.org/abs/2102.04378>
- [32] Deep Learning for Person Re-identification: A Survey (2020). <https://arxiv.org/abs/2001.04193>

Note: Sources [8] and [30] (CLIP-ReID) are cross-listed in Categories C and H, as in the original survey, since the paper is simultaneously a CLIP-adaptation study and a cross-camera generalization study. Newly added sources are indicated by 2025–2026 publication years and IEEE Xplore DOIs.